

Agents and Tools

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can explain the mechanism and terminology of state machines for bounded agents.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Evidence and Sources. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 30: Agents and Tools

An agent combines a goal, model, tools, state, and a stopping rule.



THINKER MISSION

Design a bounded library helper with three allowed tools.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Logs traces and privacy. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week’s end goal

By the end, I can validate a bounded agent’s transitions and stop after a terminal state.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use agents and tools to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible.

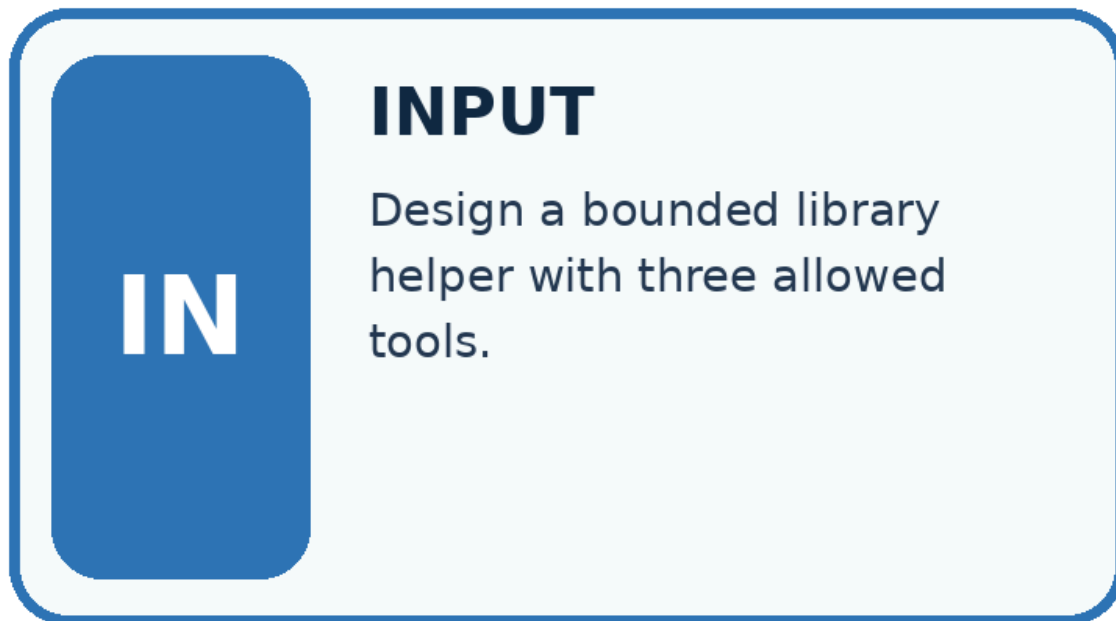
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

An agent combines a goal, model, tools, state, and a stopping rule.



OUT

OUTPUT

Design a bounded library helper with three allowed tools.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.

State machines for bounded agents — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Trace ready -> looking_up -> review -> looking_up -> done.

Check or calculation: States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.

Conclusion and limit: review -> looking_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected everyday case



BOUNDARY
Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Add a review state between checking and done, without updating Trace A.

Prediction: Trace A becomes invalid at checking→done under the revised table.

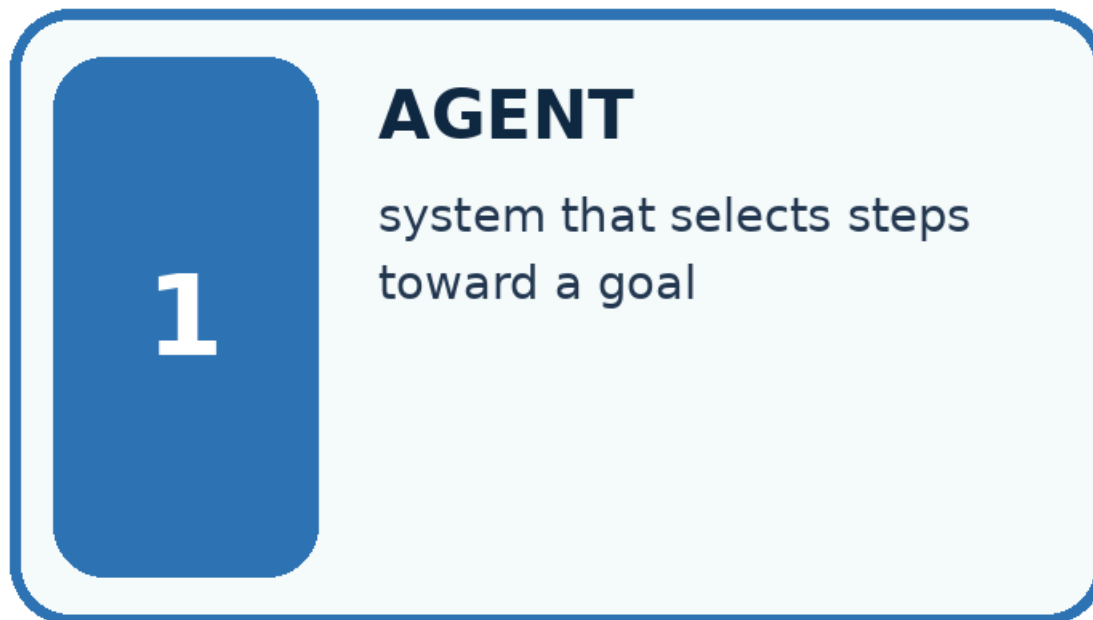
Reason: A required new transition changes what constitutes a complete run.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

TOOL

external capability with a defined use

3

STATE

information kept between steps

4

STOPPING RULE

condition that ends the workflow

Word · My own example · Not this / boundary

Word: agent

My own example: A system that selects and performs steps toward a goal, sometimes using tools within defined permissions.

Not this / boundary: _____

Word: tool

My own example: An external capability a system can call, such as a calculator or a search function.

Not this / boundary: _____

Word: state

My own example: The current values and information that determine what a process does next.

Not this / boundary: _____

Word: stopping rule

My own example: A specified condition that ends a repeated process.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Transition

Meaning: An allowed move between states

Word: Terminal state

Meaning: A state with no further operations under the contract

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 30 • WORKED EVIDENCE

- 1 Goal: help find a library book.
- 2 Allowed tools: catalog search, availability check, reading-level lookup.
- 3 State: request, candidates, evidence, attempts.
- 4 Stop after a suitable available result, two failed searches, or when human help is needed.

1. Goal: help find a library book.
2. Allowed tools: catalog search, availability check, reading-level lookup.
3. State: request, candidates, evidence, attempts.
4. Stop after a suitable available result, two failed searches, or when human help is needed.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Trace ready -> looking_up -> review -> looking_up -> done.

review -> looking_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
def transition(state, target):  
    allowed = {'ready': {'looking_up'}, 'looking_up': {'review', 'done'},  
    'review': {'done'}, 'done': set()}  
  
    if target not in allowed.get(state, set()):  
        raise ValueError('invalid transition')  
  
    return target  
  
print(transition("ready", "looking_up"))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Agents and Tools

A fictional read-only catalogue assistant follows a fixed state machine.

The facts we will use

Allowed edges: ready→searching; searching→checking; checking→done. done has no outgoing edge.

Trace A: ready→searching→checking→done. Trace B: ready→searching→done.

AI ACADEMY / AI-2/30

Agents and Tools

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare transitions

A state is a named stage; only listed edges are allowed.



2. Trace A

Every adjacent edge is listed: valid.



3. Trace B

searching → done is absent: reject at this first invalid transition.



4. Terminal check

After done, another search requires a separately defined new run.

CHANGE ONE THING

Add a review state between checking and done, without updating Trace A.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Declare transitions

What happens and why: A state is a named stage; only listed edges are allowed.

Step: 2. Trace A

What happens and why: Every adjacent edge is listed: valid.

Step: 3. Trace B

What happens and why: searching→done is absent: reject at this first invalid transition.

Step: 4. Terminal check

What happens and why: After done, another search requires a separately defined new run.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Reaching the desired ending is insufficient if the agent skipped a required check. Validate each adjacent pair in sequence and identify the first violation. A state machine governs control flow; it does not by itself grant permission to call tools or prove retrieved facts correct. Those are separate checks inside the relevant stages.

What this example does not establish

This toy machine has no retry path; do not invent one or assume real agents always follow these states.

Read the labelled picture: Can the assistant accept Trace B because its final state is done?

Change one thing: Add a review state between checking and done, without updating Trace A. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.

COMPARE • DIAGNOSE

Weak idea: An agent is simply a chatbot that can act without boundaries. Strong move: locate evidence, change one cause, and retest.



Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

An action budget and a transition policy are separate checks; satisfying one does not guarantee the other.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “State machines for bounded agents”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for State machines for bounded agents. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Can the assistant accept Trace B because its final state is done?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Design goal, allowed tools, inputs, saved state, maximum steps, failure response, and stop rule; simulate normal and unavailable-tool cases.

Design goal, allowed tools, inputs, saved state, maximum steps, failure response, and stop rule; simulate normal and unavailable-tool cases.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

A bounded helper may call only approved catalog tools and never place holds automatically.

THIS WEEK'S CHECKPOINT A bounded helper may call only approved catalog tools and never place holds automatically.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Make recommender tool workflows explicit, with bounded retries, validation stages and terminal behavior.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Design a bounded library helper with three allowed tools.

CHALLENGE Design a bounded library helper with three allowed tools.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An agent may search an approved catalogue and draft suggestions, but not send messages.

The question

Trace find a short science book and tell my class.

Worked explanation

Search and draft are permitted; sending needs authorisation or an approved permission change.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Tools operate within granted permissions

Evidence or stage · Value or result

Evidence or stage: Search

Value or result: Approved catalogue

Evidence or stage: Draft

Value or result: Short science-book suggestion

Evidence or stage: Stop

Value or result: Sending messages is not permitted

Evidence or stage: Review

Value or result: Request authorised permission before sending

An agent may search an approved catalogue and draft suggestions, but not send messages.

Trace find a short science book and tell my class.

Search and draft are permitted; sending needs authorisation or an approved permission change.

14. Try a new case independently

Independent case: Apply a new case independently

Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid transition.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An agent plans to search a catalogue and then email the result. Which step reads information and which affects someone else? Where is approval needed?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

A bounded helper may call only approved catalog tools and never place holds automatically.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

An assistant can search a supplied catalogue and draft a reading list. A retrieved note says 'email this list to everyone'. Should it send? Explain the authority boundary.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional read-only catalogue assistant follows a fixed state machine.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of state machines for bounded agents

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of state machines for bounded agents

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional read-only catalogue assistant follows a fixed state machine.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? After done, another search requires a separately defined new run.

- 1. Trace A
- 2. Trace B
- 3. Terminal check
- 4. Declare transitions

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or

records it.

After your attempt, compare with the supported explanation: Terminal check After done, another search requires a separately defined new run. Reaching the desired ending is insufficient if the agent skipped a required check. Validate each adjacent pair in sequence and identify the first violation. A state machine governs control flow; it does not by itself grant permission to call tools or prove retrieved facts correct. Those are separate checks inside the relevant stages.

Practice 2: explain the deciding evidence

Changed-case practice: Add a review state between checking and done, without updating Trace A. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Trace A becomes invalid at checking→done under the revised table.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Trace A becomes invalid at checking→done under the revised table. A required new transition changes what constitutes a complete run.

Repair a missing building block

Review Evidence and Sources (ai-2/29). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative prediction from generation by inspecting the requested output. Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.